

EDUCATION

University of Illinois Urbana-Champaign

Master of Engineering in Chemical Engineering Leadership | GPA: 3.86

Illinois, USA

Aug 2025 – Present

Nirma University

Bachelor of Technology in Chemical Engineering | GPA: 3.24

Gujarat, India

Oct 2021 – May 2025

EXPERIENCE

University of Illinois Urbana-Champaign

Graduate Research Assistant

Illinois, USA

May 2026 – Present

- Assisted in developing machine learning models to **identify safer, biodegradable alternatives to PFAS** by predicting molecular toxicity, reducing the experimental effort required for chemical screening
- Worked on a team building an end-to-end molecular machine learning workflow in PyTorch, **covering data preparation, model training, cross-validation and testing on unseen data**
- Applied **MolFormer**, a pretrained transformer model, to generate molecular representations directly from chemical structures
- Investigated PFAS-polymer interactions using molecular dynamics simulations run on the **Folding@home** distributed computing platform, evaluating polymer materials for potential **PFAS removal from wastewater**

RIPE Project @University of Illinois Urbana-Champaign

Graduate Research Assistant

Illinois, USA

Jun 2026 – Present

- Authored **8+ Standard Operating Procedures (SOPs)** and safety documentation covering autoclave operation, equipment sterilization and experimental procedures, standardizing laboratory practice across all the treatment groups
- Prepared chemical solutions for sterilizing laboratory and greenhouse equipment and carried out autoclave sterilization for a 192-plant soybean study across multiple treatment groups
- Analyzed soybean phenotyping datasets in **R** to evaluate Water Use Efficiency and prepared results for research reporting

Gujarat Narmada Valley Fertilizers & Chemicals Ltd

Process Engineering Trainee

Gujarat, India

May 2024 – Jun 2024

- Documented **SOPs** for plant shutdown, equipment isolation, depressurization, purging and lockout-tagout, and observed a scheduled maintenance shutdown of an operating plant
- Monitored operating parameters including temperature, pressure and flow using **Distributed Control Systems** under engineering supervision
- Applied **learning-from-incident** findings when documenting **SOPs**, with exposure to **HAZOP** preparation in an operating plant
- Ensured compliance with industrial safety standards through PPE adherence, hazard identification, and safe work practices in the plant

Nirma University

Researcher

Gujarat, India

Dec 2024 – May 2025

"Employment of Industrial Waste Material: Calcium Carbide Residue (CCR) as a Photocatalyst for Green Hydrogen Production"

- Filed a patent** application (pending) for a novel method using industrial waste **Calcium Carbide Residue** as a photocatalyst, achieving **~15%** higher hydrogen generation efficiency than electrolysis alone by combining electrolysis with photocatalysis
- Produced hydrogen at an applied potential of **1.3 V**, below the 1.6-2.2 V practical operating range of PEM and alkaline electrolyzers, and quantified hydrogen yield by water displacement collected at the cathode
- Deposited CCR catalyst onto electrodes by **sol-gel** method and applied a **polyvinyl alcohol** overcoat to prevent catalyst detachment during operation
- Designed a custom photo-electrocatalytic hydrogen production cell in **AutoCAD** and calculated the moles of hydrogen generated using **MS Excel**

Nirma University

Researcher

Gujarat, India

Jul 2023 – Sep 2023

"Wastewater Treatment of Dye Intermediate Industry for Zero-Liquid Discharge"

- Performed **15+** lab-scale experiments evaluating coagulant and flocculant combinations for dye-intermediate industry wastewater treatment
- Measured Chemical Oxygen Demand (**COD**) by the **closed reflux colorimetric method** and Biological Oxygen Demand (**BOD**) by the **dilution method**
- Reduced COD from the thousands of mg/L to approximately **100 mg/L** and BOD to approximately **30 mg/L** through optimized chemical dosing
- Developed and applied modified chitosan as a biodegradable flocculant

RESEARCH PROJECT

University of Illinois Urbana-Champaign

Graduate Research Project

Illinois, USA

Feb 2026 – May 2026

"AI-Assisted Physics-Guided Modelling for Carbon Capture and Gas Separation in Polymer Membranes"

- Developed predictive machine learning models, Random Forest as baseline and XGBoost as the main model, combining Morgan fingerprints with physics-based gas transport and polymer descriptors to evaluate membrane selectivity across three gas pairs for **carbon capture**
- Implemented **GroupShuffleSplit** & 5-fold GroupKFold validation grouped by polymer SMILES to ensure no data leakage
- Identified physics-based and polymer descriptors as the strongest predictors of selectivity in feature importance analysis, ranking above structural fingerprints, with higher-rigidity polymer structures associated with higher selectivity

TECHNICAL SKILLS

Laboratory & Materials: Sol-gel catalyst deposition, Electrode fabrication and coating, Photo-electrocatalytic hydrogen generation, COD and BOD analysis, Coagulation-flocculation, Modified Chitosan

Computational & Methods: MS (Excel, Word, Powerpoint), Python, R, PyMOL, Molecular dynamics simulation, AutoCAD, ChemCAD (coursework), MATLAB (coursework), Leakage-controlled model validation, Experimental design, SOP development, Six Sigma Green Belt (LinkedIn Learning)